INSTRUCTIONS

DASMixer is a de novo assistant tool for the combined analysis of tandem mass spectra, integrating the results of searches against protein and spectral databases with de novo peptide sequencing.

DASMixer includes four information modules (Figure 1):

- Samples;
- Peptides;
- Proteins;
- Reports.

Work with the software is performed sequentially through these modules.

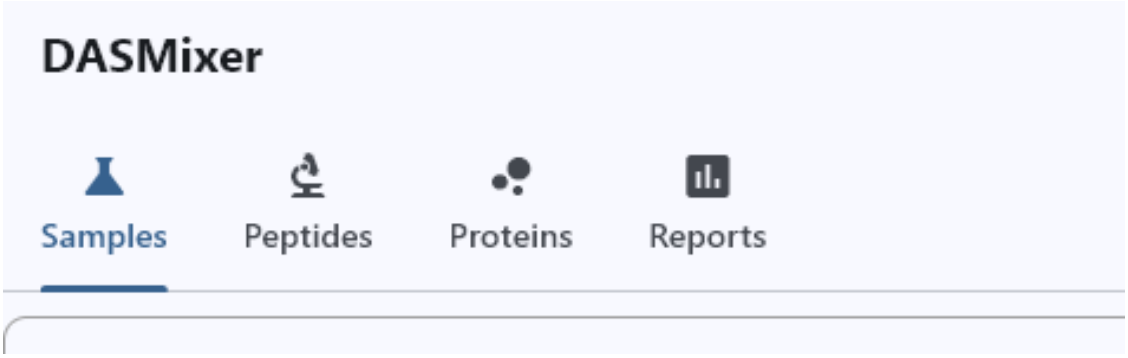


Figure 1 – DASMixer information module panel. “Samples” module blocks.

1. The “Samples” module includes four blocks::

- comparison groups (adding comparison groups);
- import spectra (importing mass spectrometry data files);
- identification tools (adding identification tools);
- import identifications (adding identification results).

1.1 In the “comparison groups” section, add a comparison group by clicking “Add group”, then specif:

— the group name;

— a short description (if necessary);

— select a color;

— click «Add».

Important: use unique group names for quick identification.

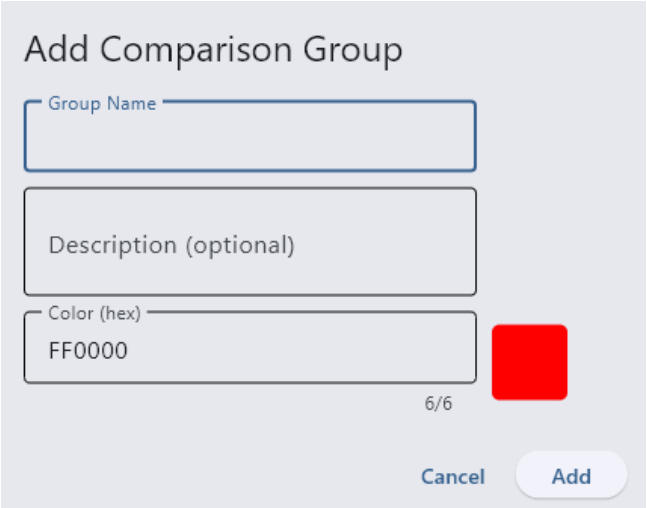


Figure 2 – Adding comparison groups.

1.2 In the “import spectra” section, upload mass spectrometry data files (spectra). Files can be uploaded in two ways (Figure 3):

— manual selection and upload of individual files using «Select files»;

— batch upload from a folder with extraction of sample IDs from file names using «Pattern Matching»;

— click «Import».

Important: verify sample IDs to ensure they are recognized correctly.

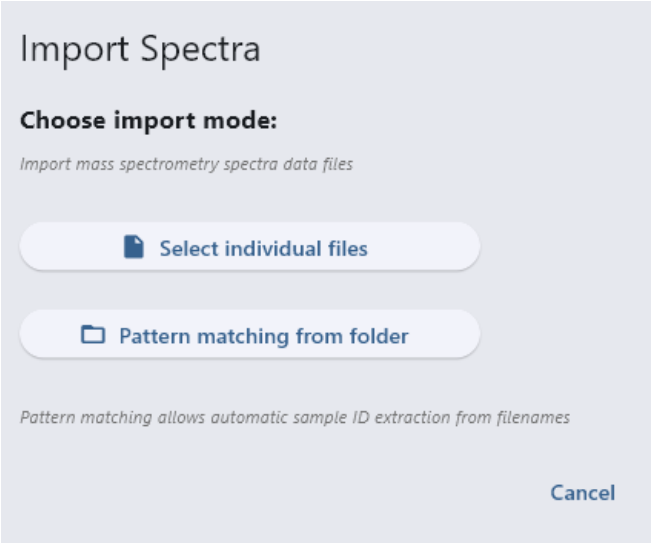


Figure 3 – Selecting the file upload mode.

1.3 In the «identification tools» block, define identification tools by first clicking «Add tool» (Figure 4):

— specify the tool name (for example, PowerNovo2, MaxQuant);

— define the identification strategy:

- library search,
- de novo sequencing;

— select a color.

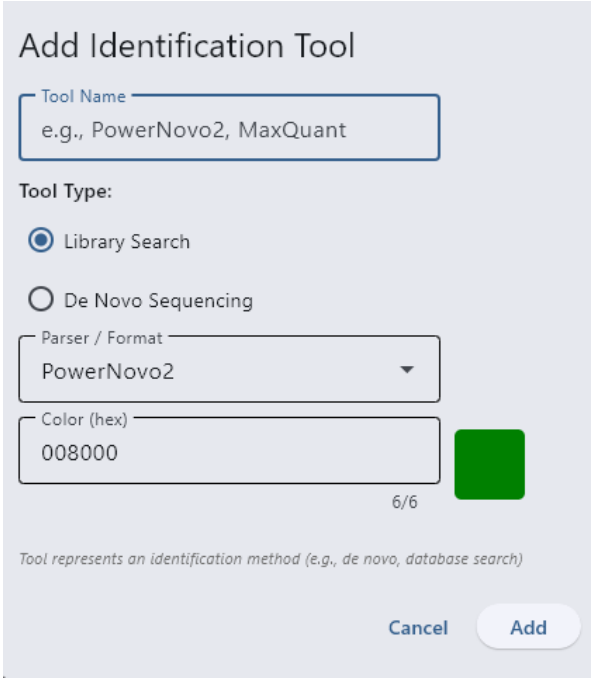


Figure 4 – Selecting identification tools.

1.4 Adding identification results in the «Import identifications» section can be performed in two modes (Figures 5,6):

– manual selection and upload of individual files using «Select files»;

– batch upload from a folder with extraction of sample IDs from file names using «Pattern Matching».

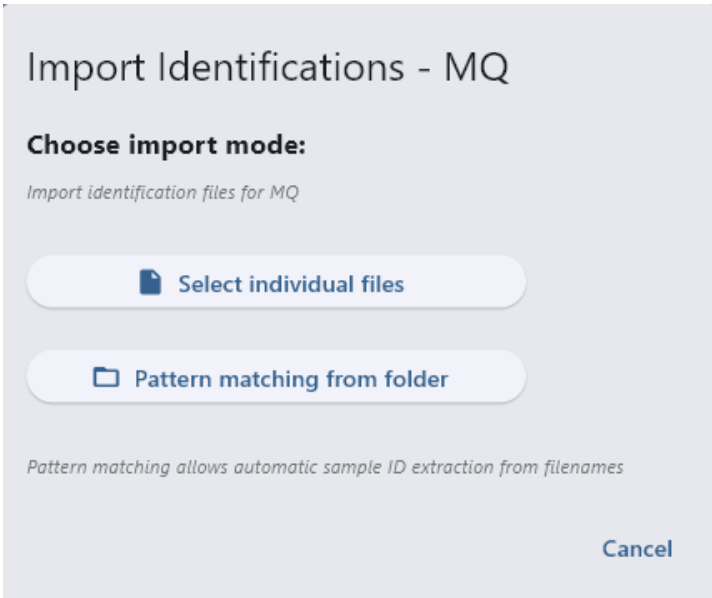


Figure 5 – Importing identification results (for MQ).

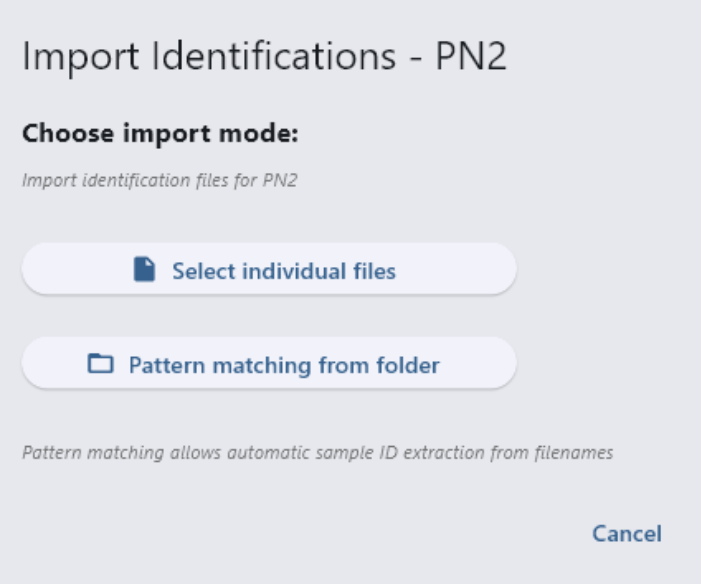


Figure 6 – Importing identification results (for PN2).

1.5 Editing sample parameters in the «Samples».

In the sample editing window (Figure 7), the following parameters can be modified:

- sample name;
- comparison group;

— additional parameters in JSON format (if necessary).

After that, save or cancel the changes.

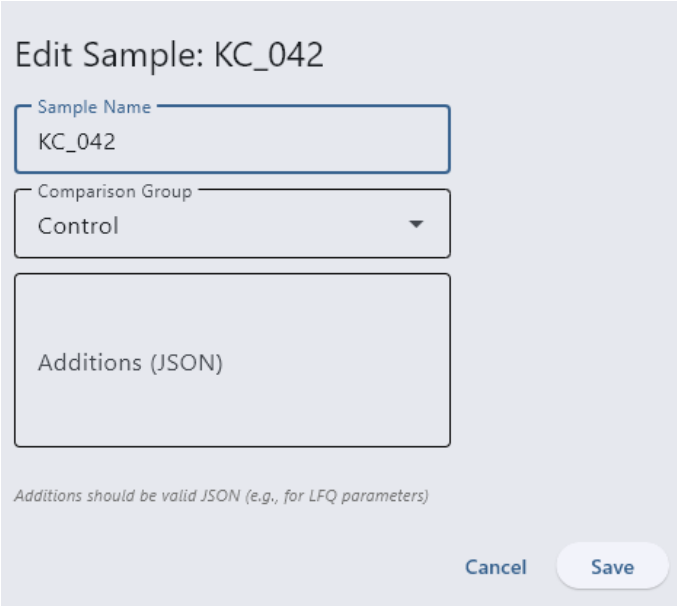


Figure 7 – Editing a sample.

2. The «**Peptides**» module includes the following seven blocks (Figure 8):
   — Protein sequence library (definition of the protein sequence library);
   — Tool settings (definition of precursor identification filtering conditions);
   — Ion matching settings (definition of filtering conditions for fragment ions);
   — Preferred identification selection (selection of the preferred identification mode);
   — Filters (filtering criteria for the resulting table);
   — Data (interactive results table);
   — Ion match plot / plot preview (graphical matching of the spectrum to fragment ions).

2.1 In the «Protein sequence library» section, upload a file containing protein sequences (Figure 8):

— click «Browse» and select a database file in *.fasta* format;

— click «Sequences in UniProt format» (if necessary), then click «Load Sequences».

Important note: the uploaded database must correspond to the organism being studied.

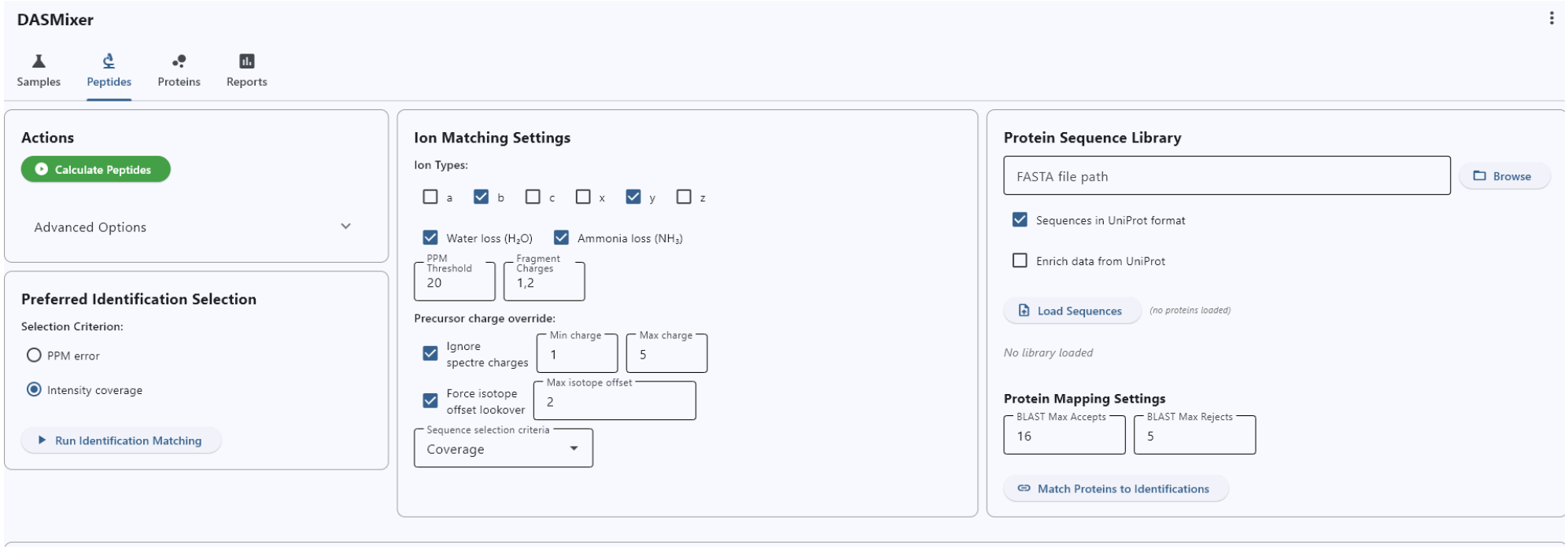


Figure 8 – «Peptides» module. Uploading a protein library.

2.2 To configure precursor ion parameters, open «Tool settings» and set the filters.

For precursor identifications:

- tolerance (allowed mass deviation);
- score (minimum score);
- ion coverage (%);
- peptide length;
- spectrum peaks (number of peaks in the spectrum).

Protein matching filters:

- protein identity;
- PTM combinations (post-translational modification combinations).

Depending on the algorithm used (Library Search or *De Novo*), additional parameters are available (Figure 9).

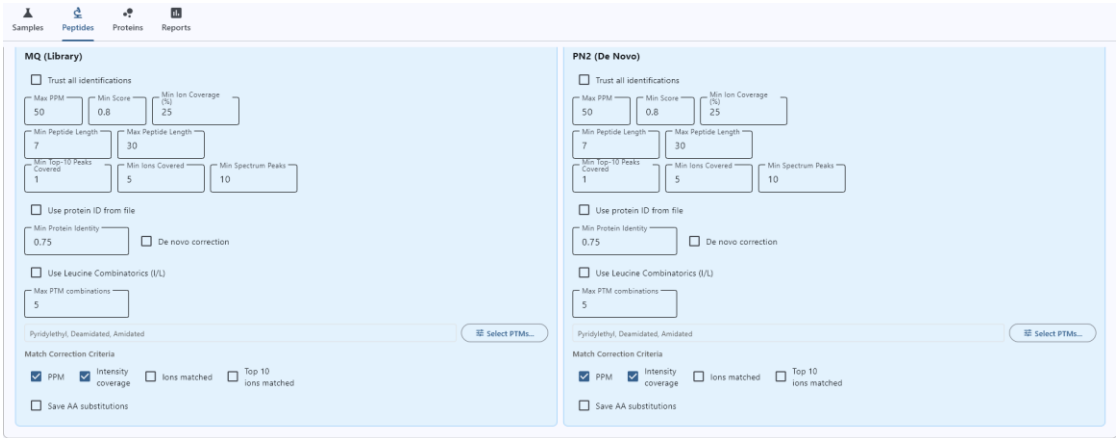


Figure 9 – Setting identification filtering parameters for Library Search and De Novo methods.

2.3 In the «Ion Matching Settings» block, select the following (Figure 10):

- ion types «Ion Types» (*a, b, c, x, y, z*);
- optionally enable water loss («Water loss») and ammonia loss («Ammonia loss»);
- set matching accuracy «PPM Threshold» (e.g., 20);
- specify allowed fragment charges «Fragment Charges» (e.g., 1,2);

In the «Precursor charge override» section:

- define the charge range (Min/Max charge);

— Ignore spectra charges (if necessary);

— Force isotope offset lookup (if necessary).

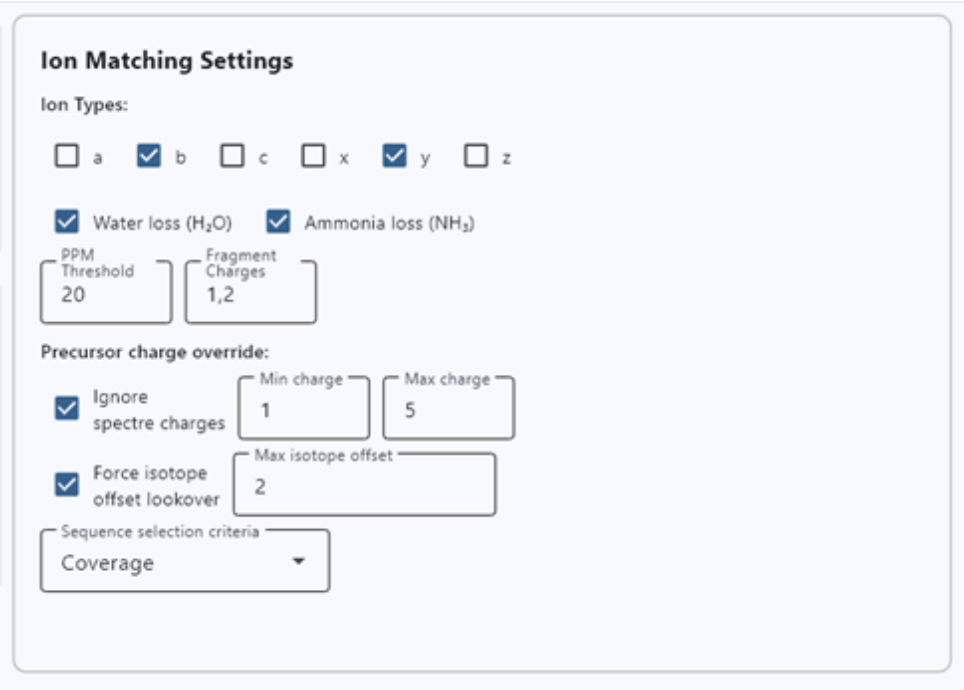


Figure 10 – «Ion Matching Settings» block.

2.4 In the «Preferred identification selection» block, choose the preferred mode for identifications (Figure 11):

— ppm error;

— intensity coverage.

After selecting the criterion, click «Run Identification Matching».

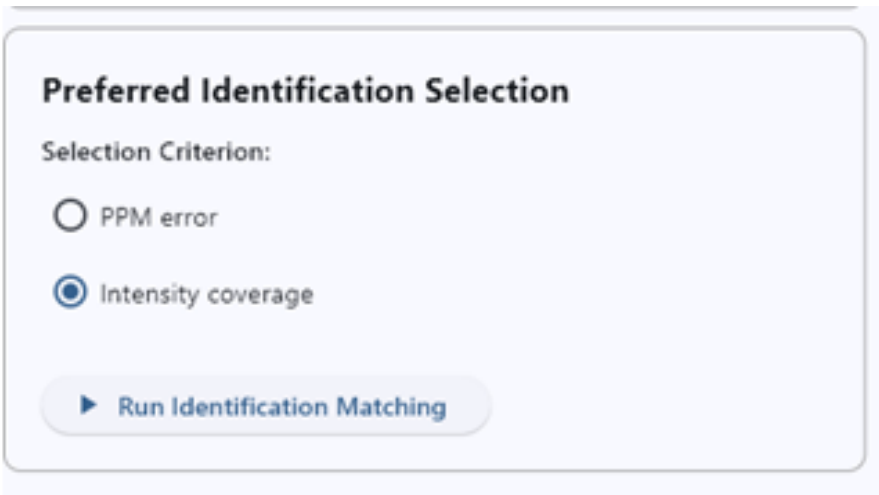


Figure 11 - Selection of identification mode.

2.5 In the «Filters» block, set filtering criteria for the resulting table.

Data can be ranked by: score, ppm, sequence, etc.

2.6 The «**Data**» block allows viewing the resulting identification table (Figure12). The table includes the following columns:

- sample;
- spectrum ID;
- identifier;
- sequence;
- PPM;
- ion type;
- ion coverage, %;
- protein;
- gene;
- plot, etc.

Samples Peptides Proteins Reports

Search and View Identifications

Peptide Identifications

Filters

Data

Sample	Spectrum ID	Seq #	Scans	Tool	ID	Sequence	PPM	Preferred	Ions Matched	Ion Type	Top-10 Peaks	Ion Coverage, %	Protein	Gene	Plot
KC_042	30	29	3869	PN2	2948	KPVEEYANC[Pyridylethyl]HLAR	1.2073		13	+y	6	30.3473			
KC_042	33	32	3942	PN2	2951	YLYENC[Carbamidomethyl]HALAR	0.4219		3	+y	3	19.9348			
KC_042	39	38	4214	PN2	2957	LVHVEEPHTETVR	0.8944		15	+y	6	36.4899			
KC_042	46	45	4504	PN2	2964	M[Oxidation]EM[Oxidation]EYLPLL...	19.2460		5	+y	3	27.7173			

Showing 1-50 of 13296 rows (Page 1 of 266)

Export

Rows per page 50

Figure 12 – Interactive results table.

2.7 The «Ion match plot / plot preview» block controls the display of graphical matching between the spectrum and fragment ions.

Display settings:

— show plot title («Show title»);

— show legend («Show legend»);

— show sequences in proteins («Show sequences in proteins»).

Note: ion matching parameters (ion types, PPM, etc.) are configured in the «Ion Settings» section (Ion Matching Settings).

After selecting the options, click «Apply Settings» to update the plot.

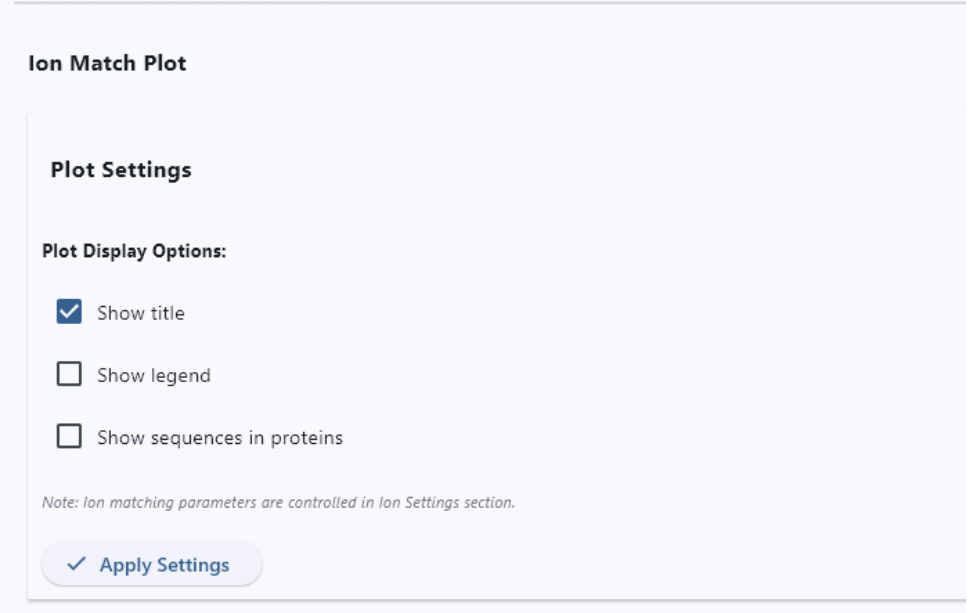


Figure 13 – Graph display settings window.

The «Plot Preview» block displays a preview of the spectrum-to-fragment-ion matching graph.

3. The «Proteins» module is designed for visualization of protein identification results and includes three blocks (Figure 14):

— Protein detection;

— Label-free quantification;

— Protein results and Data.

3.1 In the «Protein detection» section, the following parameters must be set:

— minimum number of peptides;

— minimum number of unique identifications.

3.2 In the «Label-free quantification» section, select the method:

— emPAI;

— iBAQ;

— NSAF;

— Top3;

— additional parameters.

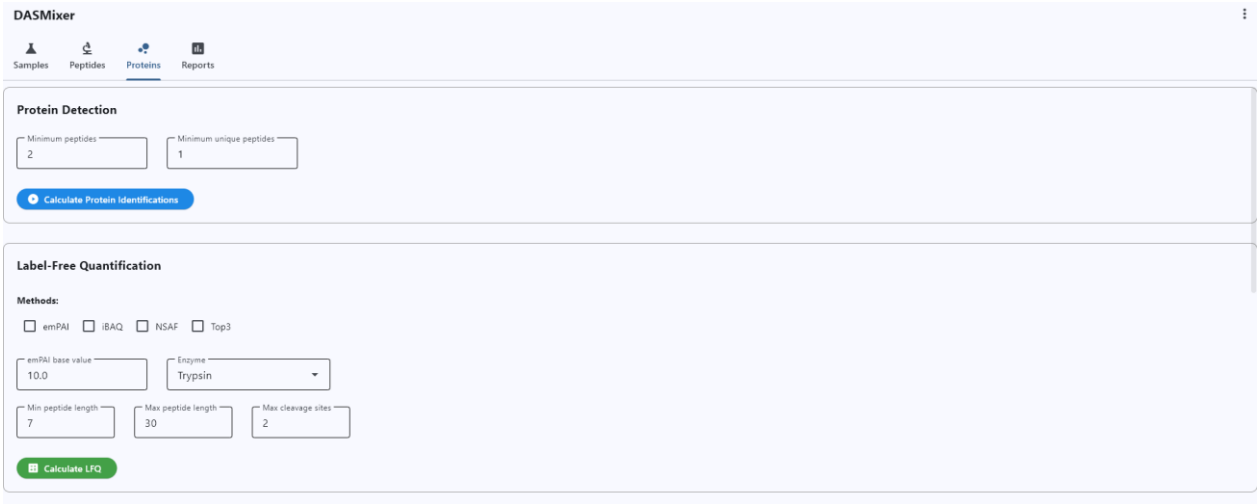


Figure 14 – Settings in the «**Proteins**» module.

3.3 In the «Protein results and Data» block, analysis results can be viewed. Filtering, sorting, and visualization tools are available.

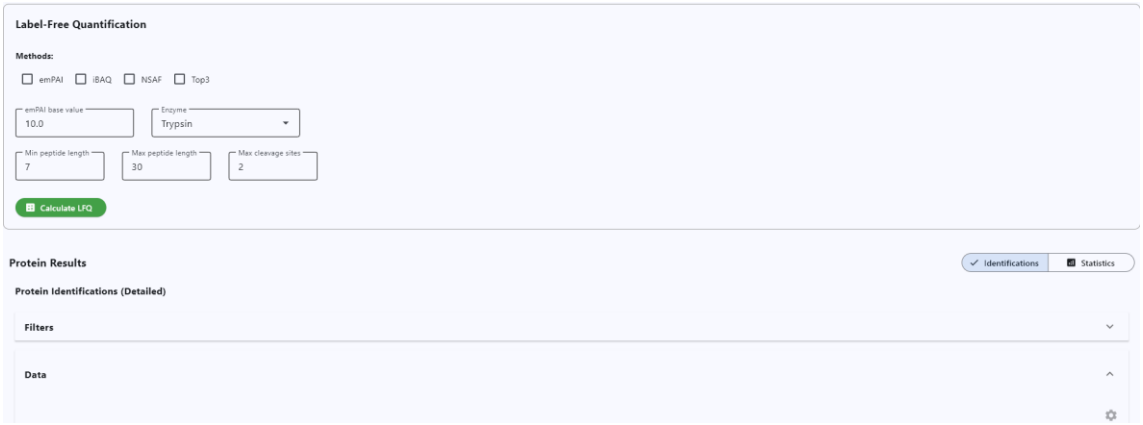


Figure 15 – «**Protein results and Data**» block.

4. The «**Reports**» module allows configuring the quality parameters of graphical output and defining the content of final reports (Figure 16).

It includes two blocks:

— Global settings;

— Available reports.

4.1 In «Global settings» define formatting requirements for figures:

— font size;

— figure size;

— click «Save Setting».

Buttons:

«Generate All Selected» – generate all selected reports;

«Export All Selected» – export all selected reports to files.

4.2 The «Available Reports» section contains report templates for analysis and result export:

— «Sample Report» (demonstration report for system familiarization);

— «Tool Match» shows an increase in the number of matches between two selected tools;

— «Volcano Report (not connected)» presents data on FC/p-value changes and volcano plot graphs.

Buttons:

— «Generate» – create a report;

— «Saved Reports» – view previously saved reports;

— «View» – open a generated report;

— «Export» – save the report to a file.

Note: In real use, «Sample Report» will be replaced with reports generated from your data.

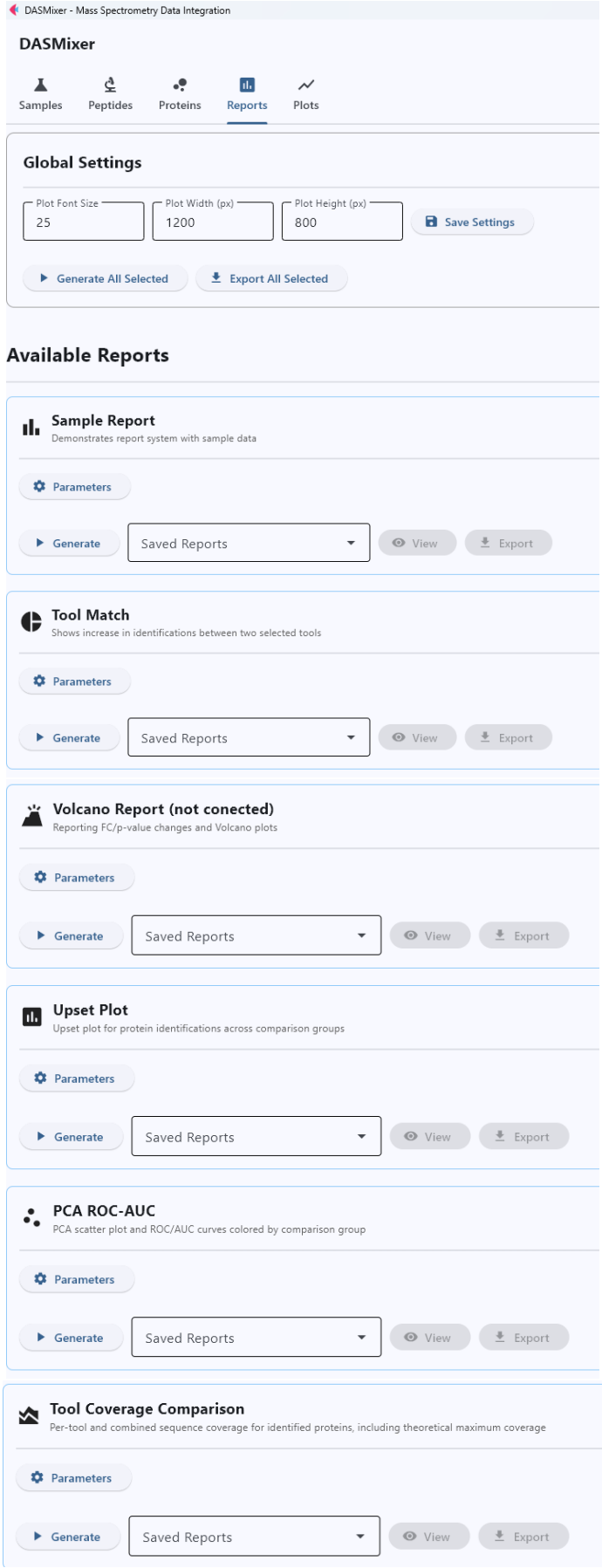


Figure 16 – Report configuration.

EXAMPLES OF REPORT MATERIALS

Interactive table of peptide identification results after combined identification using MaxQuant (MQ) and PowerNovo2 (PN2)

module "Peptides", block "Data"

DASMixer

Samples | Peptides | Proteins | Reports | Plots

Filters

Data

Sample	Spectrum ID	Seq #	Scans	Tool	ID	Sequence	PPM	Preferred	Ions Matched	Ion Type	Top-10 Peaks	Ion Coverage, %	Protein	Gene	Plot
PC_132	1687379	6367	9801	PN2	6368	LKEFGNTLEDK	4.6460		12	+y	6	33.1848	P02654	APOC1	
PC_132	1687425	6413	9879	PN2	6414	EFGNTLEDK	4.4374		18	+y	8	36.4690	P02654	APOC1	
PC_132	1687563	6551	10106	PN2	6552	EFGNTLEDK	5.3187	✓	19	+y	5	19.7410	P02654	APOC1	
PC_132	1688051	7039	10874	PN2	7040	TPDVSSALDK	5.4832	✓	11	+y	6	40.1697	P02654	APOC1	
PC_132	1688253	7241	11129	PN2	7242	TPDVSSALDK	2.2833	✓	19	+y	6	28.3497	P02654	APOC1	

Interactive figure showing the matching of the spectrum to fragment ions module "Peptides", block "Data"

Interactive table of protein identification results after combined identification using MaxQuant (MQ) and PowerNovo2 (PN2)

module “Peptides”, block “Data”

DASMixer

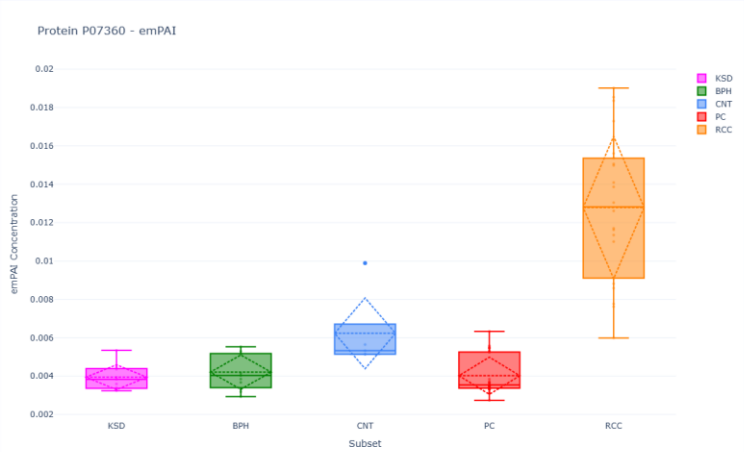
Samples Peptides Proteins Reports Plots

Protein ID	Sample	Group	Gene	Peptides	Unique Peptides	Coverage, %	Intensity Sum	emPAI	iBAQ	NSAF	Top3	MW, Da	Plot
P07360	CNT_005A	BPH	C8G	3	3	21.7822	5873944.8750	0.0052				22263.5425	
P07360	CNT_007A	KSD	C8G	2	2	14.3564	2603031.3750	0.0038				22263.5425	
P07360	CNT_010A	BPH	C8G	2	2	15.8416	1453961.8750	0.0038				22263.5425	
P07360	CNT_014A	KSD	C8G	2	2	15.8416	5271927.9375	0.0039				22263.5425	
P07360	CNT_016A	BPH	C8G	2	2	13.3663	1398155.5469	0.0032				22263.5425	
P07360	CNT_018A	KSD	C8G	3	3	21.7822	6114792.1250	0.0053				22263.5425	

Interactive figure showing protein abundance across comparison cohort samples module “Proteins”, block “Plot Preview”

DASMixer

Samples Peptides Proteins Reports Plots

Plot Settings

Plot Preview